Practicumopdracht 3 – Saving and Logging regression app Predictions

Context

This assignment is an extension of practicumopdracht 2, in which you performed a regression analysis and built a Streamlit regression app that predicts an outcome based on user input.

At the moment, the app can make predictions, but it does not yet behave like a real machine learning system.

In this practicum, you will extend your app in two ways:

- add an SQLite storage layer to log predictions
- generate synthetic input data to simulate app usage over time

Assignment

Extend your existing Streamlit regression app with an SQLite-based storage layer.

Your solution should make it possible to:

- automatically create a database if it does not yet exist
- store each prediction in a structured table withing the app
- save both the input values and the predicted output
- include a timestamp for each prediction
- Track predictions over time inside the app

In addition, generate a synthetic dataset that can be used to simulate new prediction requests.

This synthetic data should be realistic enough to resemble the original regression dataset, but should not simply be an exact copy of the original records.

Deliverables

- the updated app.py
- the SQLite database file
- the synthetic dataset you generated